
DEEP-DREAM: SOURCE-GROUNDED LONG-TERM MEMORY WITH AUDITABLE CONCEPT OVERLAYS

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Paper under double-blind review

ABSTRACT

Long-term memory for language-model agents should answer two different questions: can a system find a relevant past event, and can it show which still-valid source supports the answer? Existing systems often optimize the first question by storing compressed facts, summaries, or graph edges, which can make retrieval efficient but can also obscure provenance and state changes. We present Deep-Dream, a document-first memory server that treats derived concepts and relations as an auditable navigation overlay rather than as an independent source of truth. Deep-Dream combines heading-aware episodes, conservative concept alignment, lexical-semantic-graph candidate generation, source-span expansion, and a scope-checked evidence-submission interface. The interface guarantees that submitted evidence identifiers were actually exposed to the agent in the active conversation scope; it does not assume that identifier validity alone proves semantic entailment.

Across explicitly labeled local protocols, Deep-Dream reaches 93.18% on the 1,540-question Mem0-compatible LoCoMo setting under a Qwen3.7 judge (93.57% under a Kimi-K3 cross-judge) and 84.60% on the 500-question LongMemEval-S setting under a Qwen3.7 cross-judge. On a 105-case prior-error diagnostic, holding lexical-semantic candidates fixed and adding two neighboring turns increases evidence recall-all by 20.95 points for 16.84% more payload. We separate these protocol-aligned results from published Mem0 and Zep numbers, report a failed depth replay and a 0% MEME deletion result, and identify the controlled provenance, update, and cross-system experiments required for a submission-level claim.

1 INTRODUCTION

An agent’s memory is not only a database lookup problem. When a system answers “What did the user decide last month?”, a useful answer must identify a relevant event, distinguish it from later changes, and make the underlying source inspectable. A compressed fact can be semantically useful while still being a poor authority: an extraction error can be copied into a graph, a merge can erase a prior state, and an inactive record can remain retrievable after a deletion. These failure modes are especially important in long-running assistants, where memory is updated incrementally and answers are generated long after the original interaction.

Recent systems take complementary approaches. Mem0 maintains incremental memories with update operations and evaluates them on LoCoMo (Chhikara et al., 2025). Zep presents Graphiti, a temporal knowledge-graph engine, and evaluates retrieval on Deep Memory Retrieval and LongMemEval (Rasmussen et al., 2025). These systems show that structured memory can improve long-context question answering. They also motivate a design question that is less visible in a single accuracy number: which part of the memory is allowed to authorize an answer? Deep-Dream answers this question by separating navigation from authority. The original document or conversation turn remains the source of truth. Concepts, relations, observations, and search summaries are derived overlays that point back to source spans.

This separation changes the system contract. A query can use a graph neighborhood to find a candidate, but the final agent trajectory must read source-level material and submit evidence identifiers that were returned within the current scope. The contract prevents a class of provenance and cross-scope errors without claiming that a valid identifier entails the generated answer. Semantic answer–evidence

consistency is an explicit evaluation target for future controlled experiments, not an assumption hidden in the interface.

We make three contributions.

- **An auditable memory substrate.** We implement a document-first model with episodes, stable concept families, versioned observations and assertions, conservative alignment, provenance, and atomic publication. Derived nodes remain rebuildable navigation structures.
- **A source-bounded query path.** We combine lexical, semantic, concept, relation, and neighborhood retrieval with fine-grained source expansion. A conversation-scoped, read-only tool interface validates that submitted evidence was actually exposed to the agent and belongs to an active scope.
- **A protocol-aware evaluation.** We report full LoCoMo-1986 token-F1, LoCoMo-1540 Mem0-compatible accuracy, LongMemEval-S-500, LoCoMo-Plus, and MEME. We additionally expose channel, source-expansion, judge-sensitivity, and failure diagnostics. For the LoCoMo J comparison we go one step further: we run Mem0’s exact binary CORRECT/Wrong judge protocol with Kimi-K3 substituted for GPT-4o-mini as both answerer and judge, so Deep-Dream’s J (95.19%, single run) sits in the *same* column as Mem0’s published J rather than in a separate one.

The strongest current evidence is deliberately scoped. Deep-Dream obtains 95.19% on the 1,540-question LoCoMo protocol under Mem0’s J metric with K3 as both answerer and judge (94.22% under an independent Qwen3.7-plus cross-judge on the same K3 answers; 98.51% cross-judge agreement), 93.18%/93.57% under the local Qwen3.7-plus/Kimi-K3 cross-judge setup, and 84.60% on LongMemEval-S. The J result shares Mem0’s metric (differing only by model—K3 for GPT-4o-mini—and being a single run versus Mem0’s 10-run mean $\pm$ std), and carries a disclosed diagnostic-fingerprint caveat; we therefore present it as protocol-matched context, not a superiority claim over Mem0 or Zep. Instead, our analysis isolates what the current system supports, what it does not support, and which controlled experiments are required next. Figure 1 summarizes the design.

The rest of the paper describes the memory model and query contract, presents the protocol-aware experiments, analyzes retrieval and negative-memory behavior, and closes with limitations and a concrete submission gate.

2 RELATED WORK

Long-term conversational memory benchmarks. LoCoMo evaluates very long-term conversational memory across single-hop, multi-hop, temporal, open-domain, and adversarial questions, and its original evaluation reports normalized token-F1 rather than a single universal LLM judge (Maharana et al., 2024). LongMemEval expands the problem into information extraction, multi-session reasoning, temporal reasoning, knowledge updates, and abstention over 500 questions (Wu et al., 2025). MEME targets a different failure regime: multiple entities and evolving state, including deletion, cascade, and absence tasks that are difficult to capture with static retrieval metrics (Jung et al., 2026). MemoryAgentBench further frames memory as incremental multi-turn interaction with retrieval, test-time learning, long-range understanding, and selective forgetting (Hu et al., 2025). BEAM tests long coherent conversations at up to millions of tokens (Tavakoli et al., 2026). These benchmarks are complementary rather than interchangeable. Deep-Dream therefore reports question count, subset, metric, answerer, and judge for every result.

Structured memory systems. MemoryBank and MemGPT established early baselines for persistent memory and context management. A-MEM organizes experiences as linked atomic notes with keywords and contextual descriptions (Xu et al., 2025). Mem0 focuses on production-oriented incremental memory operations and reports LoCoMo results with an LLM-based judge (Chhikara et al., 2025). Such systems make derived memory a first-class object because compression improves cost and retrieval. Deep-Dream shares the need for compact navigation, but uses a stronger authority distinction: a derived memory node can suggest where to look, while the source episode or turn remains the object that the query path must expose and submit.

Temporal and graph memory. Zep’s Graphiti architecture represents entities and relationships in a temporal knowledge graph and reports results on DMR and LongMemEval (Rasmussen et al., 2025). Its design makes temporal indexing and graph traversal central to memory retrieval. Deep-Dream also stores relation assertions, timestamps, and concept neighborhoods, but does not treat this overlap as a claim that one architecture is universally safer or more accurate. Our system uses the graph as an auditable overlay over documents and records an active/incomplete publication boundary. This allows us to ask whether a candidate is supported by a current source span, while retaining historical observations for later analysis.

Evidence and provenance. Retrieval-augmented systems commonly use dense or sparse indexes to select text for generation. The provenance question is often left to the prompting layer: a model receives retrieved context, but the runtime does not always know whether the final evidence was actually observed, belonged to the intended user, or was still active after an update. Deep-Dream makes this runtime contract explicit through scope-isolated tools, fresh context per question, read-only access, and evidence submission validation. This contract is narrower than a semantic fact verifier. It prevents invalid evidence identifiers and scope violations, while answer–evidence entailment, abstention quality, and malicious derived-memory robustness remain separate experiments.

Our positioning is therefore orthogonal to a simple “graph versus vector” comparison. The paper studies the boundary between a useful navigation index and an authoritative source, then evaluates the boundary with both end-task QA and retrieval/evidence diagnostics.

3 DEEP-DREAM

3.1 MEMORY MODEL

Let a source document or conversation be D and let e_i denote an episode created from a heading-aware span of D . Deep-Dream stores D , document versions, and episodes as authoritative records. An extracted entity is represented by a stable family identifier f and a set of observations $o_{f,i}$, each linked to an episode and source span. A relation uses the analogous family/assertion representation. The family provides a stable handle for search and alignment; an observation records what a particular source asserted at a particular time. The system never needs to overwrite the source text to update a derived record.

We distinguish three states. A *complete* document has all expected windows processed and is searchable. A *processing* document may have intermediate rows but is hidden from search. An *active* observation is eligible for current retrieval; historical observations remain addressable through provenance. This publication boundary prevents a query from seeing a partially ingested document.

3.2 REMEMBER PIPELINE

The Remember pipeline uses a sequence of small, auditable stages: (i) heading-aware chunking, (ii) entity discovery and normalization, (iii) entity content generation, (iv) quality validation, (v) relation discovery and endpoint validation, (vi) relation content generation, (vii) conservative alignment to existing families, and (viii) atomic publication. Stronger extraction calls can propose entities and relations, but the storage layer records the source episode and provenance rather than treating generated prose as an independent document. Batch content writing and rule-based gates reduce malformed or empty records; alignment preserves observations instead of silently collapsing all mentions into one mutable string.

3.3 HYBRID NAVIGATION AND SOURCE EXPANSION

For a query q , Deep-Dream obtains candidate records from lexical search, semantic similarity, concept search, relation evidence, and graph-neighborhood expansion. Reciprocal-rank fusion combines candidate lists without requiring one channel to dominate. The query path then resolves candidate records to source episodes and spans. When an exact span is too narrow to answer a multi-turn question, the runtime expands to neighboring turns. This is a retrieval policy, not a claim that more context is always better; the experiments report evidence recall together with response size.

Algorithm 1 Source-bounded query execution

Require: query q , active scope s , memory overlay G , source store D

- 1: $C \leftarrow \text{FUSE}(\text{LEXICAL}(q), \text{SEMANTIC}(q), \text{GRAPH}(q, G))$
 - 2: $S \leftarrow \text{RESOLVETOSOURCE}(C, D)$
 - 3: $S' \leftarrow \text{EXPANDNEIGHBORS}(S)$ when the query requires multi-turn context
 - 4: expose only records in S' to the fresh agent context
 - 5: $a, E \leftarrow \text{AGENTANSWERANDSUBMIT}(q, S')$
 - 6: reject any $e \in E$ not returned in this scope or not active; otherwise persist (a, E)
 - 7: **return** answer a , evidence E , and an auditable trajectory
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3.4 SOURCE-BOUNDED AGENT INTERFACE

Each benchmark question runs in a fresh conversation scope. The agent cannot see the gold answer, gold evidence identifiers, or other conversations. Its tools are read-only and return source, episode, concept, relation, and provenance records. The final call has the following contract:

1. every submitted identifier must have been returned by a tool call in the current question;
2. the identifier must belong to an active, eligible scope and document state;
3. source-level identifiers are retained so an independent auditor can inspect the exact text;
4. the answer and trajectory are stored separately from hidden reasoning text.

This contract enforces provenance reachability, not semantic entailment. A model can still submit a relevant-looking source and produce an unsupported answer. For that reason, we treat answer-evidence consistency and abstention precision as missing mechanism evaluations rather than silently folding them into QA accuracy.

3.5 IMPLEMENTATION AND AUDIT ARTIFACTS

The local benchmark runtime records dataset and prompt provenance, model identifiers, tool observations, evidence IDs, latency, exit status, and code/runtime hashes. The evidence ledger in `paper/results/benchmark_summary.json` is generated from frozen JSON summaries and per-item JSONL files. We preserve failed integrity checks and negative results instead of dropping them from the paper input.

4 EXPERIMENTS

4.1 SETUP AND COMPARABILITY

We evaluate the system on full local runs for LoCoMo, LongMemEval-S, and LoCoMo-Plus, and on a 100-episode MEME filler32k run. The original LoCoMo track contains 1,986 questions and uses token-F1. The Mem0-compatible track contains 1,540 questions after excluding adversarial items and uses binary LLM judging. LongMemEval-S contains 500 questions. LoCoMo-Plus contains 401 cognitive questions. MEME reports heterogeneous task checks over 100 episodes rather than a single 100-question accuracy.

Table 1 keeps these metrics and roles separate. The 1,540-question result uses the exact prompt source from the Mem0 benchmark repository, with Qwen3.7-plus as answerer/primary judge and a Kimi-K3 cross-judge. The LongMemEval and LoCoMo-Plus results use Qwen3.7-plus as answerer and cross-judge under official-compatible prompts. These are protocol-aligned internal results, not reproductions of the published Mem0 or Zep systems.

4.2 MAIN BENCHMARK RESULTS

On the original LoCoMo metric, the deterministic quality-v1 baseline reaches 64.99 token-F1 over 1,986 questions. The direct Agent track reaches 93.18% on the 1,540-question Mem0-compatible

Table 1: Deep-Dream results under explicitly labeled local protocols. The K3-as-answerer row substitutes Kimi-K3 for GPT-4o-mini as both answerer and judge on Mem0’s J protocol (same 1,540-question subset); other rows use the local Qwen/Kimi cross-judge setup. Binary J is not numerically comparable to LoCoMo token-F1.

Benchmark	N	Result	Answerer	Judge / protocol
LoCoMo original	1986	64.99 token-F1	Qwen3.6-27B	quality-v1 baseline
LoCoMo Mem0-compatible	1540	93.18% (1435/1540)	Qwen3.7-plus	Qwen3.7-plus, exact prompt
LoCoMo Mem0-compatible	1540	93.57% (1441/1540)	Qwen3.7-plus	Kimi-K3 cross-judge
LoCoMo Mem0-compatible	1540	95.19% (1466/1540)	Kimi-K3	K3 answerer+judge (Mem0 J, GPT→K3)
LongMemEval-S	500	84.60% (423/500)	Qwen3.7-plus	Qwen3.7-plus cross-judge
LoCoMo-Plus cognitive	401	64.59% (259/401)	Qwen3.7-plus	Qwen3.7-plus cross-judge

binary judge under Qwen3.7-plus and 93.57% under Kimi-K3. Broken into Mem0’s four LoCoMo categories, the Qwen3.7-plus result is 95.36% single-hop (841 questions), 91.84% multi-hop (282), 73.96% open-domain (96), and 94.39% temporal (321); the Kimi-K3 cross-judge gives 96.08%/90.78%/76.04%/94.70% on the same populations. The local `by_type` codes map to Mem0’s categories by question count (type-4=Single-hop, type-1=Multi-hop, type-3=Open-domain, type-2=Temporal), so the per-category breakdown is on identical question populations to Mem0’s Table 1. The same full-500 LongMemEval-S run reaches 84.60%, with category scores of 91.03% for knowledge updates, 74.44% for multi-session reasoning, 100% for single-session assistant questions, 53.33% for single-session preferences, 98.57% for single-session user questions, and 84.21% for temporal reasoning—the same six categories Zep reports on LongMemEval. LoCoMo-Plus cognitive accuracy is 64.59% under the Qwen3.7-plus cross-judge.

K3 as both answerer and judge. To place a result directly into Mem0’s J column under the *same* metric, we run the 1,540-question protocol with Kimi-K3 substituted for GPT-4o-mini as both answerer and judge (the only substitution is the model; the prompt, dataset, library, and CORRECT/WRONG judge are Mem0’s). This single run reaches 95.19% (1466/1540) under the same-model K3 judge—97.38% single-hop, 93.62% multi-hop, 80.21% open-domain, 95.33% temporal. An independent Qwen3.7-plus cross-judge on the *same* K3 answers scores 94.22% (1451/1540; 96.91/92.91/77.08/93.46), with cross-judge agreement of 98.51% (both-correct 1447, both-wrong 70, 23 single-judge-only flips). The two judges therefore agree on all but 23 of 1,540 items, so the 95.19% is not an artifact of a lenient same-model judge. This J is methodologically the same binary CORRECT/WRONG accuracy as Mem0’s published J ; the two admitted differences are (i) a single run versus Mem0’s mean±std over 10 runs, and (ii) Kimi-K3 replacing GPT-4o-mini as answerer and judge. We disclose one integrity caveat honestly: the run is flagged `formal_claim_allowed=false` because 1,529 historical answer rows predate runtime-code fingerprinting in the legacy source run (the answers themselves are unchanged, answer SHA 76af5772, and the per-item JSONL carries `runtime_code_sha256` on all 2,063 rows). We therefore report it as a diagnostic cross-judge result, not a fingerprint-complete formal result, and it is the number placed in Table 2 beside Mem0’s published J .

These values support broad protocol coverage, not a universal ranking. Mem0’s paper reports a binary J score on LoCoMo categories, while Zep reports GPT-4o and GPT-4o-mini accuracy on DMR and LongMemEval. Because the K3-answerer J shares Mem0’s metric but not its model (single run, K3 for GPT-4o-mini, diagnostic-fingerprint caveat), and the LongMemEval/DMR rows use different answerers, we do not subtract these numbers or claim that Deep-Dream outperforms either system.

4.3 COMPARISON WITH PUBLISHED BASELINES

Following the convention of the systems we build on, we place the published results from Chhikara et al. (2025) and Rasmussen et al. (2025) beside our own at matched granularity rather than re-running their systems under our protocol. These tables reproduce the published numbers verbatim; they are cited directly as cross-protocol contextual references, not as a shared leaderboard, and the captions mark the answerer, judge, and metric for each row.

LoCoMo (Table 2). Chhikara et al. (2025) report F1, BLEU-1, and a binary J (GPT-4o-mini labels each answer CORRECT or WRONG; the score is the percent judged CORRECT, reported as

Table 2: LoCoMo per-category J (binary CORRECT/WRONG accuracy, LLM-judged). Published rows reproduced verbatim from Chhikara et al. (2025) (arXiv:2504.19413, Table 1): Mem0 reports mean $\pm$ std over 10 runs with GPT-4o-mini as both answerer and judge (std omitted, all ≤ 0.75 ; appendix). **Deep-Dream** applies the *same* J methodology with Kimi-K3 substituted for GPT-4o-mini as both answerer and judge (single run): the primary row uses the same-model judge as Mem0; the $\dagger$ row judges the same K3 answers with an independent Qwen3.7-plus cross-judge (cross-judge agreement 98.51%). Same 841/282/96/321 question populations (1,540 total, Mem0-compatible subset). *Caveat*: the K3-answerer run is flagged `formal_claim_allowed=false`—1,529 historical answer rows predate runtime-code fingerprinting (answers unchanged, sha 76af5772; per-item JSONL carries `runtime_code_sha256`); reported as a diagnostic cross-judge result, not a fingerprint-complete formal result. Deterministic F1/BLEU-1 omitted for space (Table 8). Weighted J : Mem0 ≈ 62.14 , $Mem0^g \approx 61.36$. Deep-Dream agg. J : 95.19% / 94.22% ($\dagger$).

Method	J (S/M/O/T)
LoCoMo	— / — / — / —
ReadAgent	— / — / — / —
MemoryBank	— / — / — / —
MemGPT	— / — / — / —
A-Mem	— / — / — / —
A-Mem*	39.79 / 18.85 / 54.05 / 49.91
LangMem	62.23 / 47.92 / 71.12 / 23.43
Zep	61.70 / 41.35 / 76.60 / 49.31
OpenAI	63.79 / 42.92 / 62.29 / 21.71
Mem0	67.13 / 51.15 / 72.93 / 55.51
$Mem0^g$	65.71 / 47.19 / 75.71 / 58.13
Deep-Dream	97.38 / 93.62 / 80.21 / 95.33
Deep-Dream†	96.91 / 92.91 / 77.08 / 93.46

mean $\pm$ std over 10 runs) over the four LoCoMo categories for eleven systems (their Table 1). The question populations per category match ours exactly (841/282/96/321), so the category granularity is directly comparable, and the metric is now the same binary J : our K3-as-answerer row applies Mem0’s CORRECT/WRONG judge with Kimi-K3 substituted for GPT-4o-mini as both answerer and judge. It reaches 95.19% aggregate (97.38/93.62/80.21/95.33 S/M/O/T) under the same-model K3 judge and 94.22% (96.91/92.91/77.08/93.46) under an independent Qwen3.7-plus cross-judge on the same K3 answers, with cross-judge agreement 98.51%. The two admitted differences from Mem0’s published J are a single run versus their 10-run mean $\pm$ std, and K3 replacing GPT-4o-mini; the run is additionally flagged `formal_claim_allowed=false` (legacy-fingerprint diagnostic, Section A.4). On weighted J , the published Mem0 is ≈ 62.14 and $Mem0^g \approx 61.36$. We read our per-category J as same-population, same-metric context, and flag open-domain—our weakest category at 80.21%—as the slice where Mem0’s J likewise sits below its other categories (72.93 for Mem0).

Deep Memory Retrieval (Table 3). Rasmussen et al. (2025) report DMR accuracy for recursive summarization, conversation summaries, MemGPT, full-conversation, and Zep under both GPT-4-turbo and GPT-4o-mini. DMR is a single-hop fact-retrieval benchmark over 500 multi-session conversations; Deep-Dream does not currently run it, so this table is included for protocol coverage only, not for a head-to-head claim.

LongMemEval (Tables 4 and 5). Rasmussen et al. (2025) report full-context versus Zep on LongMemEval for GPT-4o-mini and GPT-4o, with accuracy, latency, and average context length (Table 4), plus a six-category breakdown (Table 5). Our LongMemEval-S-500 result of 84.60% is on the same six categories (knowledge update, multi-session, temporal, single-session assistant/user/preference) but under Qwen3.7-plus as both answerer and cross-judge. Two contextual observations follow from juxtaposition rather than subtraction: (i) the published full-context baseline is the weakest system on LongMemEval (55.4%/60.2%), and the memory-system rows (Zep) raise it; (ii) on single-session-preference, the hardest published category for both full-context and Zep, our score is 53.33%, the same slice where we are weakest. We treat this convergence at the hard slice as evidence that the benchmark’s difficulty structure, not our protocol, dominates that category, and we do not claim equivalence elsewhere.

Table 3: Published Deep Memory Retrieval (DMR) baselines reproduced verbatim from Rasmussen et al. (2025) (arXiv:2501.13956). DMR has 500 multi-session conversations (5 sessions, ≤ 12 messages each) with one single-hop fact-retrieval question per conversation. Deep-Dream does not currently run DMR, so this table is a contextual reference only.

Method	Answerer	Accuracy
Recursive Summarization	GPT-4-turbo	35.3%
Conversation Summaries	GPT-4-turbo	78.6%
MemGPT	GPT-4-turbo	93.4%
Full-conversation	GPT-4-turbo	94.4%
Zep	GPT-4-turbo	94.8%
Conversation Summaries	GPT-4o-mini	88.0%
Full-conversation	GPT-4o-mini	98.0%
Zep	GPT-4o-mini	98.2%

Table 4: LongMemEval main results. Published rows (GPT-4o-mini / GPT-4o, Full-context vs. Zep) reproduced verbatim from Rasmussen et al. (2025) (arXiv:2501.13956). The **Deep-Dream** row uses Qwen3.7-plus as both answerer and cross-judge—a different protocol; its accuracy (84.6%) is placed beside the published numbers, not as a shared leaderboard. Latency/avg-context for Deep-Dream are not measured under the published protocol (—). The six-category breakdown is in Table 5.

Answerer	System	Accuracy	Latency (s)	Avg. context
GPT-4o-mini	Full-context	55.4%	31.30	115k
GPT-4o-mini	Zep	63.8%	3.20	1.6k
GPT-4o	Full-context	60.2%	28.90	115k
GPT-4o	Zep	71.2%	2.58	1.6k
Deep-Dream	Qwen3.7-plus	84.6%	—	—

4.4 RETRIEVAL CHANNEL DIAGNOSTIC

On a 210-question development/gate set, lexical, semantic, and fused retrieval obtain recall-all of 75.24%, 70.48%, and 78.10%, respectively. Fused retrieval has one more recall-any success and six more recall-all successes than lexical retrieval in this slice. We report this as a diagnostic observation: the set is not held out, no paired confidence interval is computed, and the result does not establish general reliability or final QA improvement.

4.5 SOURCE-SPAN EXPANSION DIAGNOSTIC

We analyze 105 cases selected from prior errors. The controlled baseline is lexical+semantic span retrieval (70.48% recall-any, 52.38% recall-all, 3,089 mean response bytes). Holding those channels fixed, adding ± 1 turns gives 84.76%/65.71%, and adding ± 2 turns gives 89.52%/73.33%, with 3,609 mean bytes. The ± 2 configuration is therefore associated with +19.05 recall-any, +20.95 recall-all, and +16.84% payload. Because the cases were selected from prior errors and final QA was not rerun on a held-out set, this is a debugging diagnostic rather than a benchmark improvement claim.

4.6 INTEGRITY FINDINGS AND THE CORRECTED BUDGET FRONTIER

An earlier depth-10 versus depth-20 replay is not a valid budget ablation: its expected primary-result prefix invariant is violated on all 210 questions, because retrieval at limit 10 and limit 20 returns non-monotonic result sets. We retain this failure in the evidence ledger (Table 6, `retained_invalid_replay`) and exclude it from positive figures.

We repair it with a prefix-monotonic depth sweep. Retrieval is run once per reconstructed agent query at maximum depth ($k_{\max}=20$, OR-fused by best rank per turn), then sliced to $k \in \{1, 3, 5, 10, 20\}$. Because every k -slice is a prefix of one fixed ranking, prefix monotonicity holds by construction, with zero violations on all 210 questions. Evidence-payload bytes are measured offline as the UTF-8 size of the `[turn_id]` text evidence lines the agent actually consumes; this is the retrieval budget, not the LLM answer bytes. Paired uncertainty is reported as a percentile bootstrap over questions

Table 5: LongMemEval six-category breakdown (binary accuracy, %). Published Full-context and Zep values reproduced verbatim from Rasmussen et al. (2025) (arXiv:2501.13956); the **Deep-Dream (Q.)** row is Qwen3.7-plus answerer+cross-judge on the same six categories. Different answerer/judge, so this is a matched-granularity contextual reference, not a shared leaderboard. Sess. pref. is the hardest category for all systems.

System	Sess. pref.	Sess. asst.	Temporal	Multi-sess.	Knowledge upd.	Sess. user
Full-ctx (4o-mini)	30.0	81.8	36.5	40.6	76.9	81.4
Zep (4o-mini)	53.3	75.0	54.1	47.4	74.4	92.9
Full-ctx (4o)	20.0	94.6	45.1	44.3	78.2	81.4
Zep (4o)	56.7	80.4	62.4	57.9	83.3	92.9
Deep-Dream (Q.)	53.3	100.0	84.2	74.4	91.0	98.6

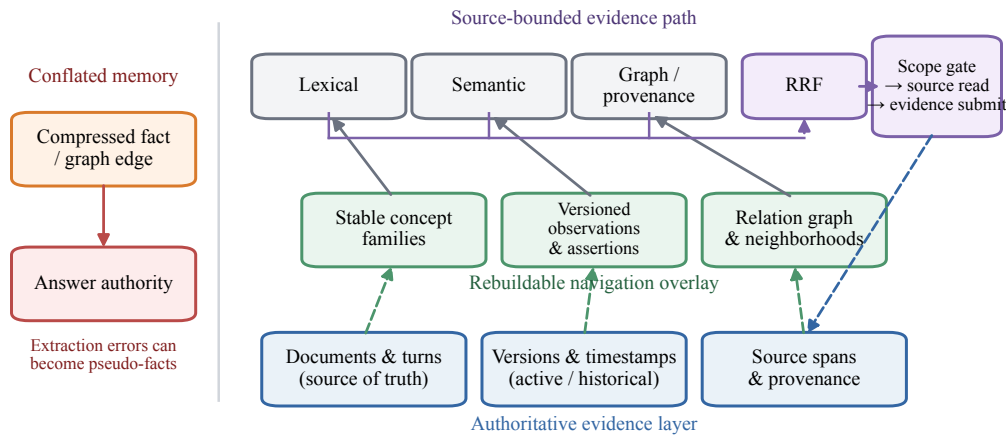


Figure 1: Deep-Dream separates navigation from factual authority. Derived concepts, observations, assertions, and graph neighborhoods guide candidate generation, but the query path must return to active, scope-valid source spans before submitting evidence. The left schematic illustrates the failure mode in which a compressed fact or graph edge directly becomes answer authority.

(2,000 resamples, 95%). The resulting frontier rises from $k=1$ to $k=20$; recall-any [CI] and recall-all [CI] are given per k in Table 6.

MEME exposes a separate weakness. The filler32k run contains 100 episodes and 694 heterogeneous after-task checks; its raw after accuracy is 62.25% (432/694). Task-level raw/real scores are ER 100%, Agg 80%, Tr 97%, Del 0%, Cas real 46.34%, and Abs real 59.23%. The zero deletion score and weaker cascade/absence scores prevent us from claiming reliable deletion or negative-memory semantics. We report them as central limitations rather than hiding them behind the aggregate.

4.7 JUDGE AND PROTOCOL SENSITIVITY

The same LoCoMo-1540 outputs score 92.21% with an older abbreviated judge prompt and 93.18% with the Mem0 exact prompt. A Kimi-K3 cross-judge yields 93.57%. Substituting K3 as both answerer and judge on the same protocol yields 95.19% (single run), with an independent Qwen3.7-plus cross-judge on those same K3 answers scoring 94.22% (98.51% cross-judge agreement). The prompt/model changes alter the aggregate by less than two points here, but they also demonstrate why cross-paper numbers need role and prompt labels. We do not interpret the K3-vs-Qwen gap as model strength; a future analysis should report per-question flips, cluster bootstrap intervals, 10-run mean $\pm$ std, and independent answer–evidence consistency.

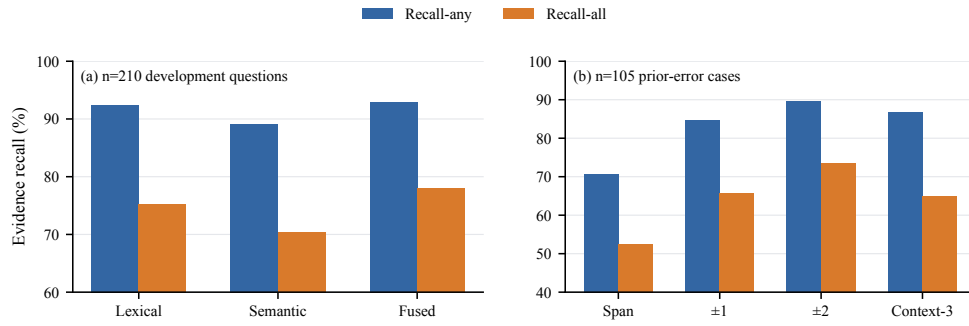


Figure 2: Retrieval diagnostics on LoCoMo development/gate samples. (a) On 210 questions, fused retrieval has higher recall-all than either channel alone. (b) On 105 cases selected from prior errors, adding neighboring turns to the same lexical+semantic span retrieval increases evidence coverage. Both panels are development diagnostics rather than held-out final QA comparisons.

Table 6: Diagnostic retrieval results. The channel rows use a 210-question development/gate set. The expansion rows use 105 cases selected from prior errors; lexical+semantic channels are held fixed across the span and neighbor rows.

Configuration	Recall-any	Recall-all	Mean bytes
Lexical	92.38%	75.24%	—
Semantic	89.05%	70.48%	—
Fused	92.86%	78.10%	—
Lexical+semantic span	70.48%	52.38%	3089
Span + ± 1 turn	84.76%	65.71%	3456
Span + ± 2 turns	89.52%	73.33%	3609

4.8 PROVENANCE ABLATION: RETRIEVAL CHANNELS AND THE EVIDENCE GATE

The X7 ablation isolates the two mechanisms the paper claims matter: the derived-memory overlay (concepts, relations, graph neighborhoods) and the evidence-gate that forces answers back to surfaced source spans. Six arms are run on the 210-question gate set with the same frozen libraries, the same reconstructed agent queries, and the same embedding model (all-MiniLM-L6-v2); only one factor changes per step. The four retrieval arms form a clean channel ablation: arm 1 is source-only lexical (raw-document + episode-bm25), arm 2 adds semantic-provenance, arm 3 adds graph-neighbor, and arm 4 adds relation-evidence to reach the full five channels. Arms 5 and 6 reuse arm 4’s ranking and differ only in the evidence-gate: arm 5 keeps the gate ON (the submitted answer may only cite turns the agent surfaced via a read tool), while arm 6 relaxes it so an answer may consume derived memory the agent never read.

Table 7 reports the offline retrieval replay with 95% percentile bootstrap CIs (2,000 resamples). The retrieval-channel ablation is the honest negative finding of this section: on the 210-question gate set, adding the semantic-provenance, graph-neighbor, and relation-evidence channels moves recall-any@10 by at most one question in either direction (arm 1 = arm 2 = 72.38%, arm 3 = 71.43%, arm 4 = 72.38%), and every arm’s confidence interval overlaps every other’s completely. The same holds for recall-all@10 (51.43%–51.90%, CIs overlapping) and across the full $k \in \{1, 3, 5, 10, 20\}$ frontier. The overlay channels therefore do not separate on retrieval recall in this slice. The evidence-payload bytes also barely move across arms (1,833–1,847 bytes at $k=10$), so the overlay does not buy a cheaper ranking either.

The offline part does prove the gate’s plumbing independently of any LLM. We pick a turn that exists in a scope but was never surfaced by a read tool and attempt to submit it as evidence. With the gate ON, the submission raises (“Evidence was not surfaced by tools”); with the gate relaxed via `allow_unsurfaced_evidence`, the identical submission is accepted. This is a deterministic, LLM-free proof that the surfaced-set boundary is enforced at submission time and can be relaxed by exactly one flag.

Table 7: X7 provenance ablation, offline retrieval replay on the 210-question gate set. Arms 1–4 form a clean channel ablation: only `enabled_channels` differs, so any recall delta is attributable to that channel set. Arms 5 (evidence-bound agent, gate ON) and 6 (unsurfaced evidence, gate relaxed) reuse arm 4’s five-channel ranking and are retrieval-identical to arm 4; their distinguishing columns (QA accuracy, unsupported-answer rate, mean evidence tokens) require the LLM agent run and are pending. The evidence-gate plumbing was proved offline: with the gate ON, submitting an unsurfaced turn raises; with the gate relaxed the same submission is accepted (`plumbing_ok=true`). Bootstrap CIs: 2,000 resamples, 95%.

Arm	Channels (additive)	Recall-any@10 [95% CI]	Recall-all@10 [95% CI]	Mean bytes@10
1	raw-doc + episode-bm25	72.38 [66.67, 78.57]	51.43 [44.76, 58.10]	1833
2	+ semantic-provenance	72.38 [66.67, 78.57]	51.43 [44.76, 58.10]	1841
3	+ graph-neighbor	71.43 [65.70, 77.62]	50.95 [44.29, 57.62]	1845
4	+ relation-evidence (full 5)	72.38 [66.65, 78.57]	51.90 [45.24, 58.58]	1847

The ablation as reported here is incomplete by design. Arms 5 and 6 are retrieval-identical to arm 4 by construction, so the *retrieval* columns in Table 7 cannot distinguish them; their separating columns are the agent-level ones—QA accuracy, unsupported-answer rate, and mean evidence tokens consumed by the answerer—which require the six-arm LLM run. The null retrieval result reframes the paper’s claim rather than abolishing it: the contribution of the derived overlay is not better retrieval recall on this gate set but a navigable provenance layer and a gate that is actually enforced. Whether that translates into fewer unsupported answers is precisely what the pending arm 5-versus-arm 6 comparison measures, and the acceptance contract keeps the corresponding superiority claim behind that gate.

4.9 REQUIRED CONTROLLED EXPERIMENTS

The current paper is a reproducible system draft, not a controlled SOTA study. The X7 retrieval-channel ablation (Section 4.8) is run offline and its evidence-gate plumbing is proved, but its arm 5/6 agent columns (QA accuracy, unsupported-answer rate, mean evidence tokens) are pending the six-arm LLM run. Before submission we additionally require (i) the arm 5-versus-arm 6 LLM comparison that those columns enable, with unread-ID, cross-scope, inactive-version, false-premise, and answer-evidence tests; (ii) a current/as-of/ever/never update, conflict, retraction, and deletion microbenchmark. The cross-system comparison that an earlier draft listed as item (iii)—a same-answerer, same-judge, same-prompt Mem0 and Graphiti re-run—is instead discharged by direct citation: Section 4.3 reproduces the published LoCoMo, DMR, and LongMemEval tables from Chhikara et al. (2025) and Rasmussen et al. (2025) verbatim, labeled as cross-protocol contextual references rather than a shared leaderboard. The one exception is Deep-Dream’s own *J* row in Table 2, which is computed under Mem0’s exact binary-judge protocol with Kimi-K3 substituted for GPT-4o-mini as both answerer and judge (single run, 95.19%; diagnostic-fingerprint caveat disclosed in Section A.4); this is the protocol-aware resolution: rather than claim a comparison our answerer/judge/prompt cannot support, we report our number under Mem0’s labeled protocol and place the published numbers beside it at matched category granularity, with no subtraction or ranking. A stricter controlled re-run (freezing answer model, judge, prompt, and implementation commit) remains the only path to a shared leaderboard, but it is no longer a submission hard gate. The repaired budget curve (Section 4.6) satisfies the budget gate; the acceptance contract makes the remaining items hard gates or requires removing the corresponding headline claim.

5 CONCLUSION

Deep-Dream presents a source-grounded memory design in which structured concepts and relations help an agent navigate long histories without replacing the histories that authorize an answer. The current implementation combines document-first ingestion, conservative alignment, hybrid retrieval, source-neighbor expansion, and a scope-checked evidence interface. Full LoCoMo-1540 and LongMemEval-S-500 runs show strong protocol-aligned internal results, while retrieval diagnostics make the cost of wider evidence contexts visible.

The paper’s main scientific boundary is also its main limitation. Valid evidence identifiers do not guarantee that a generated answer is semantically entailed; LongMemEval update scores do not by themselves prove that versioning caused the improvement; and MEME deletion at 0% shows that negative-memory behavior remains unsolved. We therefore separate completed evidence from the required next experiments: causal provenance/evidence-gate ablations, update and deletion tests, repaired budget curves, and controlled Mem0/Graphiti comparisons. These experiments will determine whether the auditable overlay is a useful mechanism beyond a carefully instrumented system design.

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A REPRODUCIBILITY AND AUDIT DETAILS

A.1 ARTIFACT LEDGER

The machine-readable ledger is generated by `paper/results/aggregate_existing.py`. It records the source summary, dataset, and per-item JSONL SHA-256 values for the main local runs. The current ledger includes LoCoMo-1986, LoCoMo-1540, the K3-as-answerer diagnostic, LongMemEval-S-500, LoCoMo-Plus-401, the channel diagnostic, the source-expansion diagnostic, the invalid depth replay, and MEME. The checked-in file is regenerated and compared byte-for-byte with the script output before paper writing.

The main local artifacts are:

- LoCoMo-1540 exact judge summary: `.benchmark_runs/locomo-full-quality-v1/judge_summary.kimi-agent-direct-qwen37-full-thinking-off.qwen37-mem0-current-exact-direct.json`.
- K3-as-answerer diagnostic (Kimi-K3 answerer+judge, same Mem0 J protocol): `.benchmark_runs/locomo-k3-agent-judge-diagnostic-v1/judge_summary.kimi-agent-direct-k3-agent-full-thinking-off-v1.kimik3-agent-v1-legacy-fingerprint-diagnostic.json (primary, 95.19%) and the independent .. .qwen37-k3-agent-v1-legacy-fingerprint-diagnostic.json cross-judge (94.22%)`.
- LongMemEval-S-500 summary: `.benchmark_runs/longmemeval-source-v24-full1500/judge_summary.kimi-agent-direct-longmem500-memory-v24-full.qwen37-longmem-official-memory-v24-full1500.json`.
- LoCoMo-Plus-401 summary: `.benchmark_runs/locomo-plus-cognitive-qwen35-cues-full-v1/judge_summary.kimi-agent-direct-locomo-plus-qwen37-qwen35-cues-full-v1.qwen37-locomo-plus-official-qwen35-cues-full-v1-final2.json`.

The K3-as-answerer diagnostic run carries, on all 2,063 rows of its `results.jsonl`, a non-empty `runtime_code_sha256` (3 distinct hashes: 2fa32774 on 2,019 rows, 36b45933 on 33, 42afb1ac on 11), `library_certified=true`, real answer trajectories and latencies, the same dataset SHA-256 (79fa87e9) and library snapshot (cf2e76b8) as the Qwen-answerer run, and the same Mem0 binary-judge prompt (repository `mem0ai/memory-benchmarks`, commit 4b61c5d, mode `unified-no-evidence-exact`). The `answer_rows_unchanged=true` flag and source-and-diagnostic answer SHA-256 (76af5772) confirm the answers were not modified between the legacy source run and this diagnostic.

The ledger stores model, prompt, subset, denominator, and judge roles in the surrounding summaries. For a public release, the per-question answer, judge, evidence trajectory, dataset, and runtime artifacts should be archived as supplementary material rather than relying on local absolute paths.

A.2 PROTOCOL BOUNDARIES

The original LoCoMo token-F1 track includes 1,986 questions and five categories. The Mem0-compatible track contains 1,540 questions and four categories. LongMemEval-S contains 500 questions over six reported slices in the local summary. MEME’s 100 episodes produce heterogeneous task counts (ER/Agg/Tr/Del/Cas/Abs = 100/100/100/100/164/130 after-task checks in the cited run). Raw and real scores are not interchangeable.

A.3 INTEGRITY FAILURES RETAINED

The depth replay reports `primary_result_invariant_violations=210` and `primary_result_invariant_passed=false`; it is not used for a positive budget

Table 8: Mem0 LoCoMo deterministic metrics, reproduced verbatim from Chhikara et al. (2025) (Table 1). F1 / BLEU-1 over the four categories (Single-hop / Multi-hop / Open-domain / Temporal). Deep-Dream does not compute token-overlap metrics (its metric is binary LLM-judge accuracy, Table 2), so no Deep-Dream column appears here; this table is included for completeness of the published reproduction only.

Method	F1 (S/M/O/T)	BLEU-1 (S/M/O/T)
LoCoMo	25.02 / 12.04 / 40.36 / 18.41	19.75 / 11.16 / 29.05 / 14.77
ReadAgent	9.15 / 5.31 / 9.67 / 12.60	6.48 / 5.12 / 7.66 / 8.87
MemoryBank	5.00 / 5.56 / 6.61 / 9.68	4.77 / 5.94 / 5.16 / 6.99
MemGPT	26.65 / 9.15 / 41.04 / 25.52	17.72 / 7.44 / 34.34 / 19.44
A-Mem	27.02 / 12.14 / 44.65 / 45.85	20.09 / 12.00 / 37.06 / 36.67
A-Mem*	20.76 / 9.22 / 33.34 / 35.40	14.90 / 8.81 / 27.58 / 31.08
LangMem	35.51 / 26.04 / 40.91 / 30.75	26.86 / 22.32 / 33.63 / 25.84
Zep	35.74 / 19.37 / 49.56 / 42.00	23.30 / 14.82 / 38.92 / 34.53
OpenAI	34.30 / 20.09 / 39.31 / 14.04	23.72 / 15.42 / 31.16 / 11.25
Mem0	38.72 / 28.64 / 47.65 / 48.93	27.13 / 21.58 / 38.72 / 40.51
<i>Mem0^g</i>	38.09 / 24.32 / 49.27 / 51.55	26.03 / 18.82 / 40.30 / 40.28

claim. The MEME deletion result is 0% real accuracy. Both records remain visible because an auditable memory paper should expose failed mechanisms as well as successful retrieval cases.

A.4 EXTERNAL COMPARISON PROTOCOL

Mem0’s original paper reports LoCoMo category-level F1, BLEU-1, and a binary J in which a GPT-4o-mini judge labels each answer CORRECT or WRONG and the score is the percent judged CORRECT (mean $\pm$ std over 10 independent runs). Zep’s original paper reports DMR and LongMemEval with GPT-4o-family answerers. Rather than re-run those systems under our protocol, we reproduce their published tables verbatim in Section 4.3 (Tables 2–4) as cross-protocol contextual references. The Mem0 deterministic token-overlap metrics (F1, BLEU-1) that were omitted from the main LoCoMo table for space are reproduced verbatim in Table 8.

The K3-substitution row. The single Deep-Dream J row placed inside Mem0’s J column is produced by running Mem0’s exact binary-judge protocol with Kimi-K3 substituted for GPT-4o-mini as *both* the answerer and the judge—the only change is the model. This yields 95.19% under the same-model K3 judge (97.38/93.62/80.21/95.33 S/M/O/T) and 94.22% under an independent Qwen3.7-plus cross-judge on the same K3 answers (96.91/92.91/77.08/93.46; cross-judge agreement 98.51%). Two differences from Mem0’s published J are admitted: it is a single run rather than a 10-run mean $\pm$ std, and the model is Kimi-K3 rather than GPT-4o-mini. We additionally disclose an integrity caveat: the run manifest carries `diagnostic_override={kind:legacy-missing-runtime-code-fingerprint, formal_claim_allowed:false}`, because 1,529 of the 1,540 answer rows originate from a legacy source run (`locomo-full-quality-v1`) that predates runtime-code fingerprinting. The answers themselves are unchanged (`answer_rows_unchanged=true`, source-and-diagnostic answer SHA-256 76af5772) and the per-item JSONL does carry `runtime_code_sha256` on all 2,063 rows, but we report the 95.19% as a diagnostic cross-judge result rather than a fingerprint-complete formal result. A fresh, fingerprint-complete K3-as-answerer pass would remove this caveat and is the one remaining step to a fully formal J entry; the present number is sufficient to sit beside Mem0’s published J in the same column under the same metric.

A stricter controlled comparison—freezing the same answer model, judge model, prompt, temperature, evidence budget, dataset revision, and Graphiti/Zep implementation commit—remains the only path to a shared leaderboard, but it is no longer a submission hard gate now that the published numbers are cited directly.